Software Requirement Specifications

<Project Title>

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Summery

The **Smart Warning Advisor** is a web-based system designed to help students who are on academic warning improve their CGPA. Many students face difficulty in choosing the

right courses after receiving a warning, which can further affect their academic performance.

This system analyzes the student's academic record (transcript) and recommends suitable courses based on their past performance. It also suggests expected grades, ensures that the selected courses do not clash in schedule, and keeps the total credit hours within the allowed limit.

The main goal of the system is to guide students in making better academic decisions without relying completely on manual advisors. It helps reduce confusion, improves planning, and supports students in achieving academic success.

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1. Introduction

1.1. Purpose

The purpose of the **Smart Warning Advisor** is to assist students who are on academic warning in improving their academic performance. The system helps students select suitable courses based on their previous results, so they can increase their CGPA effectively.

It aims to reduce confusion in course selection by providing smart recommendations, expected grade outcomes, and a clash-free timetable. The system also ensures that students stay within the allowed credit limits.

Overall, the purpose of this project is to provide an automated academic advising system that supports students in making better decisions and helps them achieve academic stability and success

1.2. Scope

The scope of the **Smart Warning Advisor** is limited to providing academic support to students who are on first or second academic warning. The system focuses on analyzing student transcripts and recommending suitable courses to help improve CGPA.

It includes features such as course recommendation, expected grade prediction, credit hour management, and clash-free timetable generation. The system is designed as a web-based application that can be used by students and administrators within a university.

However, the system is limited to academic advising only and does not handle full university management tasks like admissions, fee processing, or attendance management. Its recommendations depend on the accuracy of the available student data.

1.3. Product Perspective

The **Smart Warning Advisor** is a web-based application that works as a digital academic advising system. It is designed to assist students who are on academic warning by analyzing their academic records and providing intelligent course recommendations.

This system can be considered as a standalone application, but it can also be integrated with the university's existing database system to fetch student transcripts and course details. It interacts with users (students and admin) through a user-friendly interface and processes data using a backend system.

The product replaces the traditional manual advising process by providing automated, consistent, and quick recommendations. It acts as a support tool for both students and academic advisors, improving decision-making and reducing workload.

1.4. User Characteristics

The users of the **Smart Warning Advisor** system mainly include students and administrators. Their characteristics are as follows:

1. Students

- Basic knowledge of computers and internet usage
- Able to use web applications (login, upload, view results)
- May have limited technical skills
- Need guidance in course selection
- Belong to academic warning category

2. Administrator

- Good knowledge of computer systems
- Familiar with student academic records and university policies
- Responsible for managing data (students, courses, credit limits)
- Can handle system updates and monitoring

3. Technical Skills Level

- **Students:** Beginner to intermediate
- **Admin:** Intermediate to advanced

1.5. Similar apps and systems/Literature Review

Many academic advising and course recommendation systems have been developed to help students choose suitable courses and improve their academic performance.

Academic Advising Systems (AAS) are widely used in universities to guide students in course selection, academic planning, and career development. These systems help

students understand degree requirements and make better decisions. However, traditional advising systems depend heavily on human advisors, who often face workload issues and cannot give enough time to each student .

Modern systems use **recommender system techniques** such as:

- Collaborative filtering
- Content-based filtering
- Machine learning algorithms

These techniques provide personalized course suggestions based on student data and past performance. Course recommender systems have become important because students often struggle to select the right courses due to a large number of options and lack of information .

Some systems also use **artificial intelligence and data mining** to predict student grades and recommend courses accordingly. These intelligent systems improve decision-making and help students plan their academic paths more effectively .

Despite these advancements, existing systems still have limitations such as:

- Lack of real-time data integration
- Difficulty handling new students (cold start problem)
- Limited personalization in some systems

The proposed **Smart Warning Advisor** improves upon existing systems by focusing specifically on students on academic warning. It provides course recommendations, expected grade predictions, credit management, and clash-free schedules, making it more targeted and useful for struggling students.

1.5.2 systems/Literature Review

Various systems have been developed to support students in academic planning and course selection. Traditional academic advising systems mainly depend on human advisors who guide students based on their experience. While effective, these systems can be time-consuming and may not provide equal attention to all students.

Modern academic systems use **recommender system techniques** to provide automated suggestions. These include content-based and collaborative filtering methods, which analyze student data and past performance to recommend suitable courses. Such systems help students make better academic decisions and reduce confusion.

Some advanced systems also use **machine learning and data mining** to predict student performance and suggest courses accordingly. These intelligent systems improve accuracy and personalization in recommendations.

However, many existing systems have limitations, such as lack of focus on academically weak students, limited personalization, and absence of clash-free scheduling features.

The proposed **Smart Warning Advisor** addresses these gaps by specifically targeting students on academic warning. It provides personalized course recommendations, predicts expected grades, manages credit limits, and ensures clash-free schedules, making it more effective and practical for improving student performance.

1.6. Proposed Technologies

Author will provide the list of technologies which be use to develop this system for example if it is a web-based system then web associated technologies can be mention (React, Django, MySQL etc.)

2. Requirements

4. Functional Requirements

ID	Requirement	Priority	Source Use Case
FR-01	The system shall authenticate students using university Roll No and password.	High	UC-01
FR-02	The system shall fetch and parse student transcripts from the university database.	High	UC-02
FR-03	The system shall calculate the current CGPA from the parsed transcript.	High	UC-02
FR-04	The system shall identify and flag students on First or Second Warning.	High	UC-02
FR-05	The system shall generate a ranked list of course recommendations.	High	UC-03
FR-06	Recommendations shall target a minimum CGPA of 2.0 after completion.	High	UC-03
FR-07	Recommendations shall not exceed the university credit-hour limit per semester.	High	UC-03, UC-05
FR-08	The system shall detect and exclude	High	UC-05

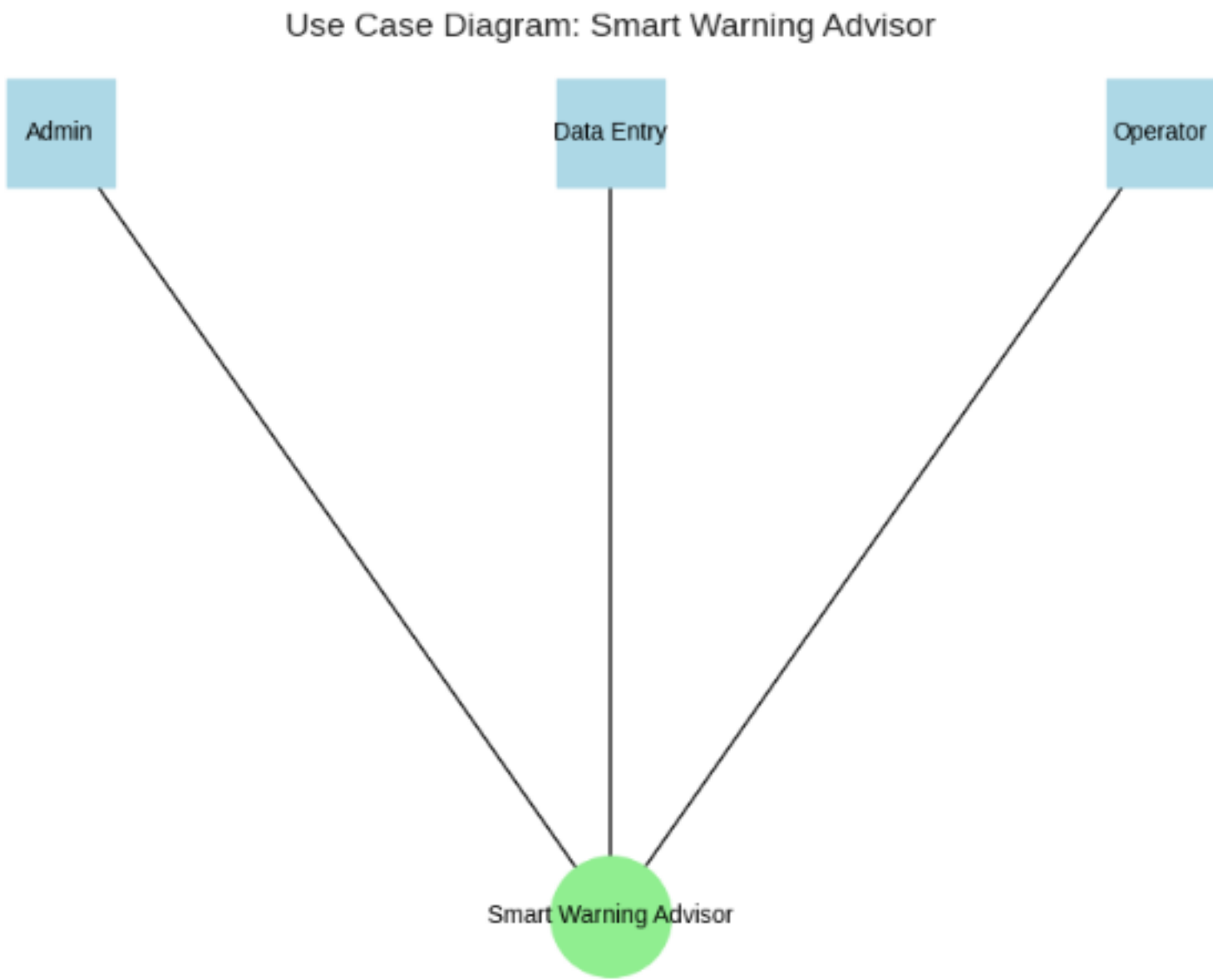
	courses with schedule conflicts.		
FR-09	The system shall display expected grade projections for each recommended course.	High	UC-04
FR-10	CGPA projections shall update in real time as the student modifies course selection.	Medium	UC-04
FR-11	Students shall be able to confirm and save their course plan.	Medium	UC-06
FR-12	Advisors shall be able to review, modify, and approve student plans.	Medium	UC-07
FR-13	The system shall send notifications on plan confirmation and advisor approval.	Medium	UC-06, UC-07
FR-14	The system shall generate downloadable reports (PDF / Excel) for advisors.	Low	UC-08

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	Recommendations shall be generated within 5 seconds for any student profile.
NFR-02	Usability	The interface shall be usable without training; all primary tasks completable in ≤ 3 clicks.
NFR-03	Security	All student data shall be transmitted over HTTPS and stored in encrypted form.
NFR-04	Scalability	System shall support concurrent use by at least 500 students without degradation.
NFR-05	Availability	System shall maintain 99% uptime during semester enrolment periods.
NFR-06	Accuracy	CGPA calculations shall be accurate to two decimal places, consistent with university records.
NFR-07	Maintainability	Codebase shall follow Django best practices and include API documentation.

3. Use Cases and Flow of Processes[a]

Figure 1: System Level Use Case Diagram



3.1. Use Case 1

Use Case ID	UC002
Name	Smart Warning Advisor
Description	This use case describes how the system provides smart warnings/ advisories to guide actors when incorrect or invalid inputs are detected during sign-up.
Requirement(s)	FR002
Actor(s)	Admin, Data Entry, Operator
Precondition	Actor attempts to sign up but provides invalid or incomplete information.
Postcondition	Actor receives advisory messages highlighting errors and suggestions for correction before proceeding.
Basic Flow	1. Actor enters credentials. 2. System validates inputs. 3. If invalid,

	system generates smart advisory messages. 4. Actor corrects inputs based on advisory. 5. On successful validation, system proceeds with account creation.
Alternative Flow	1. Admin signs in. 2. Admin creates a new user. 3. If invalid data is entered, system provides advisory messages to Admin for correction.
Exceptions	1. Invalid Inputs (e.g., wrong email format, weak password, incorrect phone number). 2. System unable to respond (technical error).
Smart Advisory Examples	"Email format is invalid. Please use example@domain.com." - "Password must be at least 8 characters including a number." - "Phone number must contain exactly 11 digits."

Author should provide the Activity diagram of each Use case. If required then sequency diagrams also add to this section.

6. End-to-End Process Flow

The following steps describe the complete flow a warning student experiences from login to enrolment approval.

1. Student opens the SWA portal and logs in with university credentials (UC-01).
2. System validates credentials and checks warning status from the university database.
3. System imports and analyses the student's academic transcript (UC-02), computing current CGPA and identifying deficient courses.
4. The recommendation engine filters the course catalogue, applies credit-limit rules, removes schedule conflicts, and ranks suitable courses (UC-03, UC-05).
5. The student views the recommended courses with projected grades and CGPA impact (UC-04).
6. The student adjusts the course selection; CGPA projection updates in real time.
7. Once satisfied, the student confirms the plan; system validates credit limit and saves the plan (UC-06).
8. The assigned academic advisor is notified; advisor reviews, optionally modifies, and approves or rejects the plan (UC-07).
9. Student receives a notification of the advisor's decision and proceeds to formal university enrolment.
10. Administrators and advisors generate reports on warning cohort progress (UC-08).

7. Constraints & Assumptions

- The university database exposes a readable API or database connection for transcript retrieval.
- Course timetable data is available in machine-readable format (JSON / CSV) before recommendation generation.
- Credit-limit rules are configurable by the administrator and may differ per programme.
- The system is intended for use during the course enrolment window only; it is not a real-time enrolment system.
- Expected grade projections are based on historical performance data and should be treated as estimates, not guarantees.

8. Glossary

Term	Definition
CGPA	Cumulative Grade Point Average — a weighted average of all course grades.
Academic Warning	Status assigned to students whose CGPA falls below 2.0.
First Warning	Issued after the first semester in which CGPA < 2.0.
Second Warning	Issued after two consecutive semesters with CGPA < 2.0.
Credit Limit	Maximum credit hours a student may register for in a single semester, per university policy.
Schedule Conflict	A situation where two courses share an overlapping time slot, preventing simultaneous enrolment.
SWA	Smart Warning Advisor — the system described in this document.